IOT ELECTIVE PITCH SESSION

Medical Clinic Digital Twin

BLE-Based Real-Time Patient Tracking & Workflow Analytics

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AGENDA

01. Introduction & Problem Statement

02. Literature Review (Related Work)

03. Existing Solutions & Their Limitations

04. Proposed "Central Link" Solution

05. Architecture & Localization Engine

06. Digital Twin Dashboard Demo

07. Implementation Plan & Hardware Progress

08. Challenges, References & Q&A

THE PROBLEM: OPERATIONAL BLINDNESS

Fragmented Patient Tracking

Medical staff currently cannot monitor precise patient movement between reception, diagnostic labs, and consultation rooms.

- **Inefficient Flow:** Waiting rooms overflow while staff are underutilized.
- **High PTAT:** Long Patient Turnaround Time negatively impacts health outcomes.
- **Static Data:** Manual check-ins miss real-time bottlenecks.

LITERATURE REVIEW (RELATED WORK)

Optimization of PTAT via BLE-Based RTLS

Arumugam & Muthaiyah (2024) developed a proof-of-concept utilizing Bluetooth Low Energy (BLE) beacons to optimize Patient Turnaround Time in public hospitals.

- **Outcome:** Their system successfully reduced total waiting times by **33% to 35%**.
- **Method:** Real-time localization using BLE signals to identify when a patient enters specific hospital zones.

Citation: Arumugam, G. R., & Muthaiyah, S. (2024). A Proof of Concept Using BLE... IJISRT, 9(5).

EXISTING SOLUTIONS & SOURCED LIMITATIONS

Current Limitations

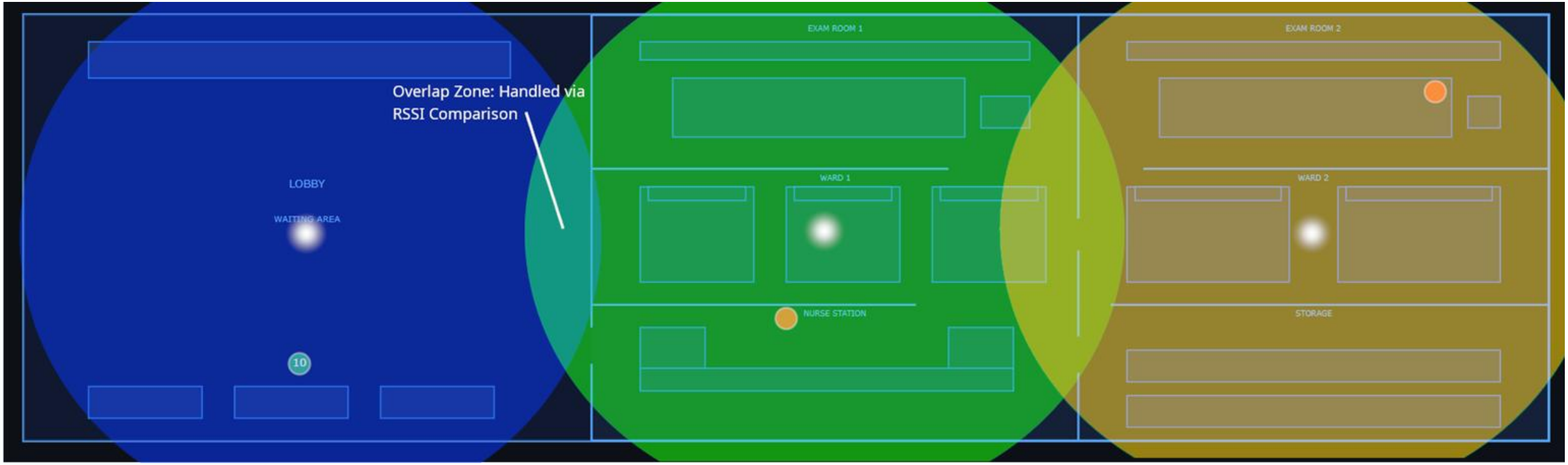
- **Manual Tracking:** High human error (22.5% inaccuracy) and paperwork overhead. *(Abdulmalek et al., 2022)*
- **RFID (Passive):** Prohibitive deployment costs and extremely short detection range. *(Yao et al., 2010)*
- **GPS/UWB:** Ineffective indoors due to signal attenuation through ceilings and walls. *(Ullah et al., 2025)*
- **WiFi:** High computational and energy costs for battery-constrained wearables. *(Hailu et al., 2025)*

Our "Central Link" Solution

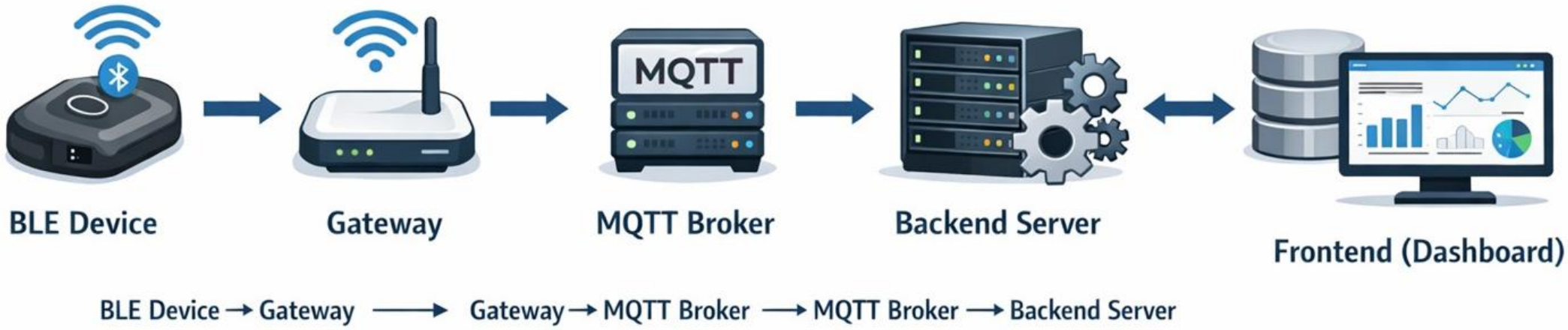
- **Live Digital Twin:** Resolves "operational blindness" via a 2D blueprint map.
- **Secure MQTTS:** Mandatory TLS 1.2/1.3 encryption for HIPAA-grade privacy.
- **AI Anomaly Detection:** Proactive identification of loitering using Isolation Forest models.
- **Scalability:** Low-power consumption suitable for lightweight clinical wearable tags.

THE DIGITAL TWIN MAPPING ENGINE

- **2D Blueprint Translation:** Translating raw gateway data into a live architectural floor plan.
- **Room-Level Accuracy:** Pinpointing patients to specific wards, exam rooms, or the lobby based on nearest-node proximity.



HIGH-LEVEL SYSTEM ARCHITECTURE



PROPOSED SOLUTION: CENTRAL LINK

Precision Edge

ESP32 Gateways scanning BLE advertising packets, ensuring low power consumption for wearable patient tags.

Secure MQTTS

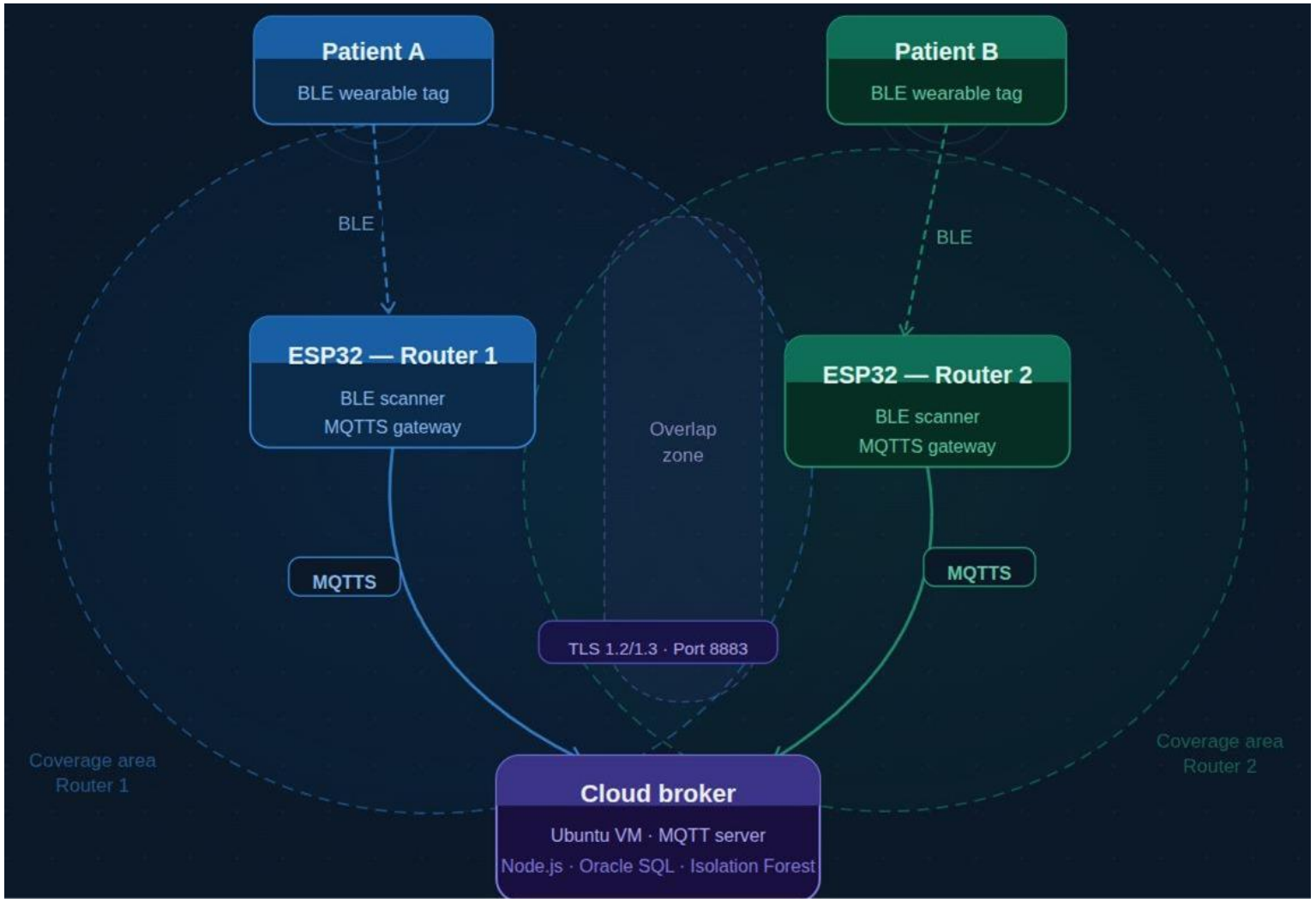
Telemetry data is encrypted using dedicated SSL certificates before reaching our Ubuntu-based cloud VM.

Predictive Analytics

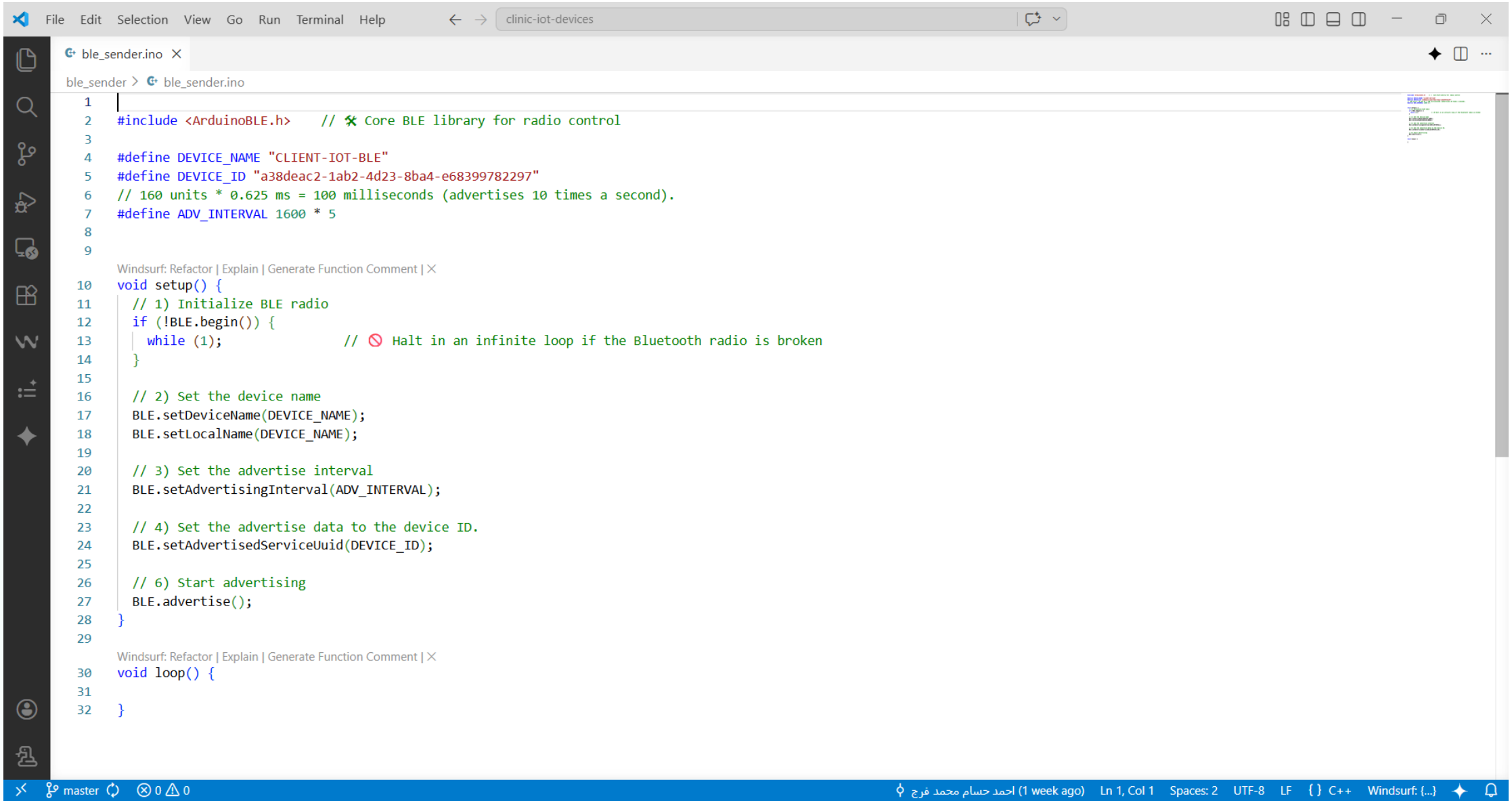
Node.js backend processes data for a Digital Twin dashboard with live heatmaps and bottleneck.

ARCHITECTURE DEEP DIVE & ZONE RESOLUTION

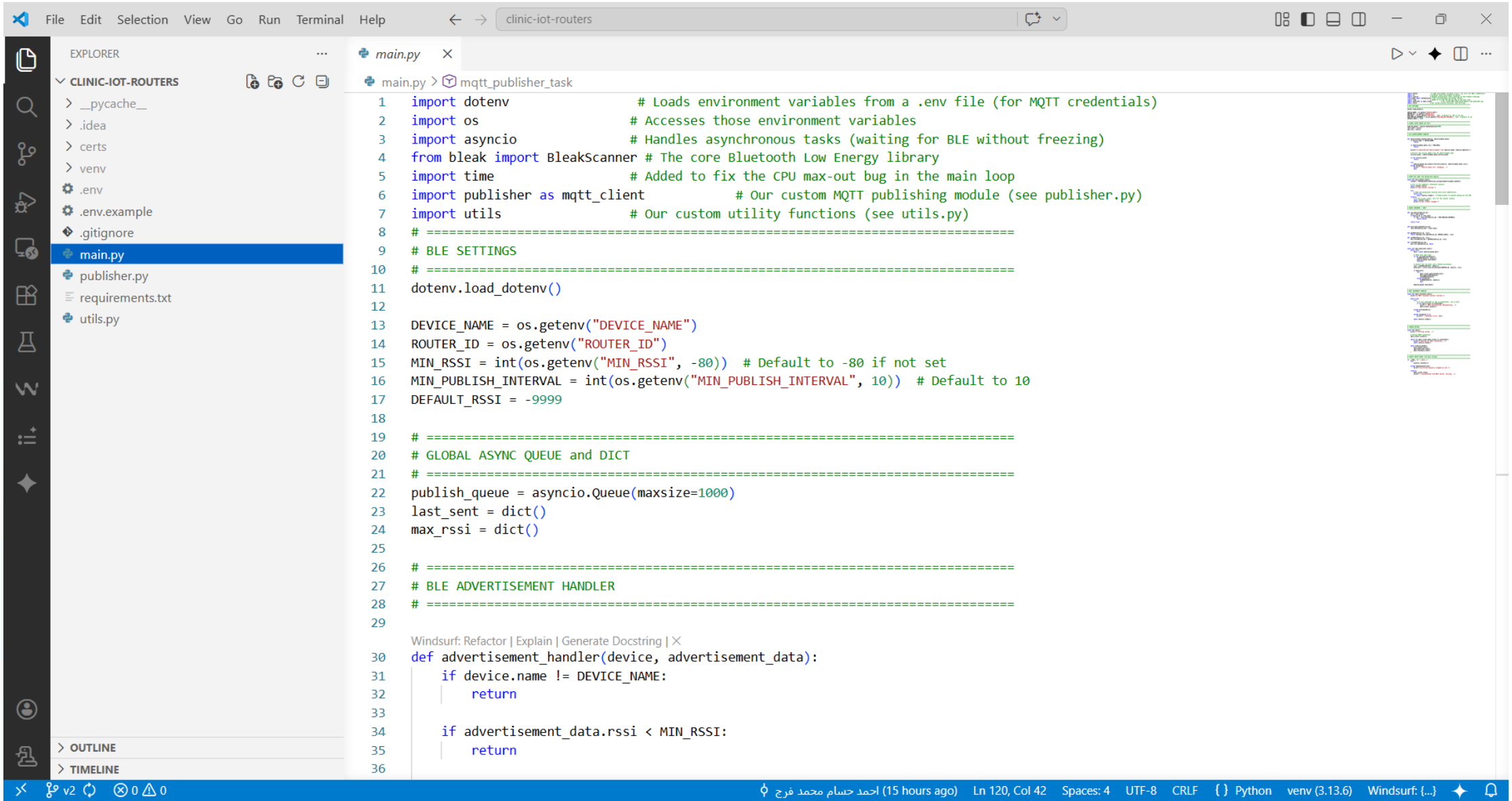
- **Secure Telemetry:** Data routed through Port 8883 using mandatory TLS 1.2/1.3 encryption.
- **Zone Resolution:** Handling overlapping gateway coverage areas by comparing real-time RSSI signal strength.
- **Cloud Processing:** Ubuntu VM hosting the Node.js backend, Oracle SQL database, and ML anomaly models.



Devices Setup



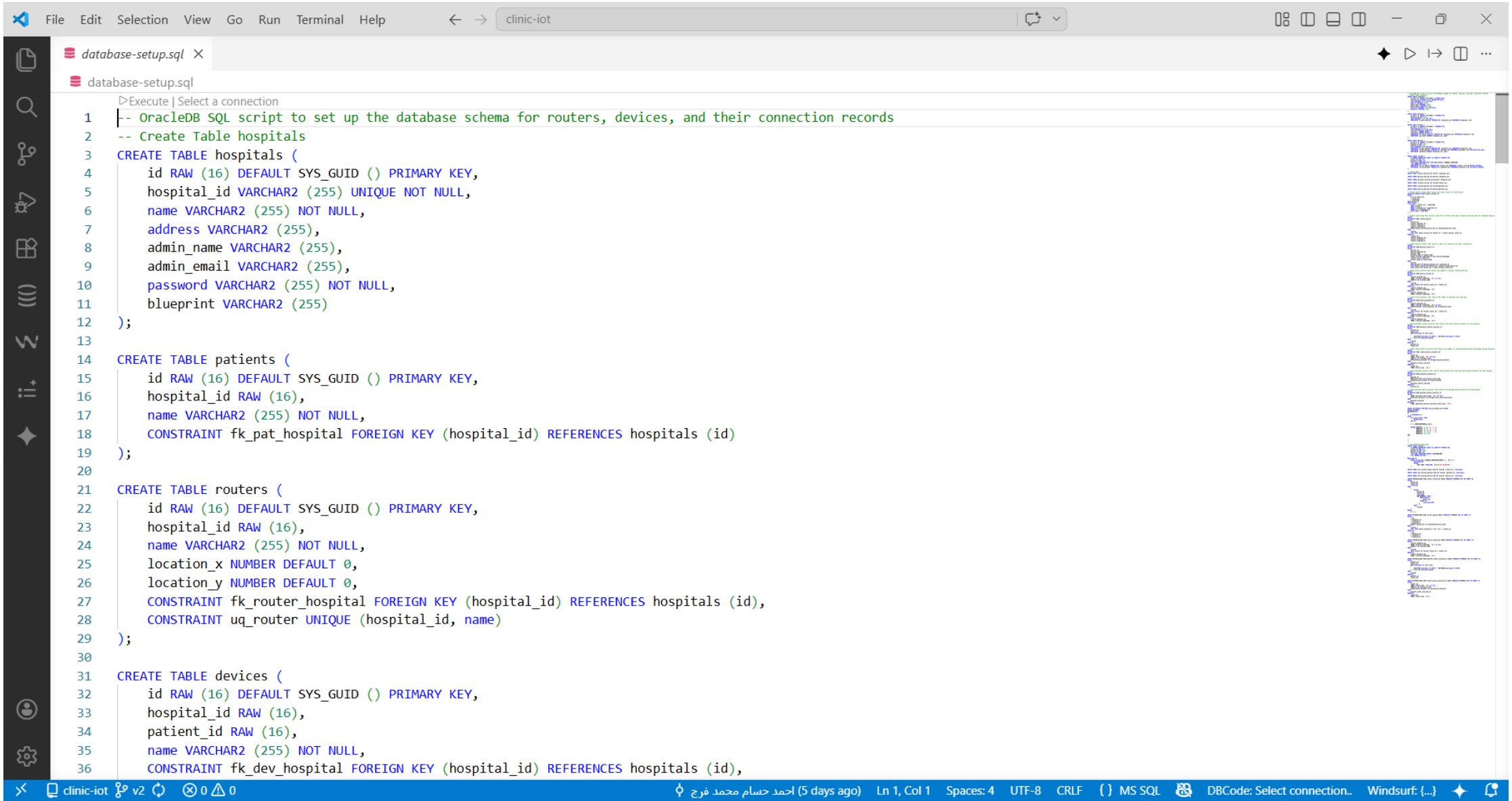
Routers Setup



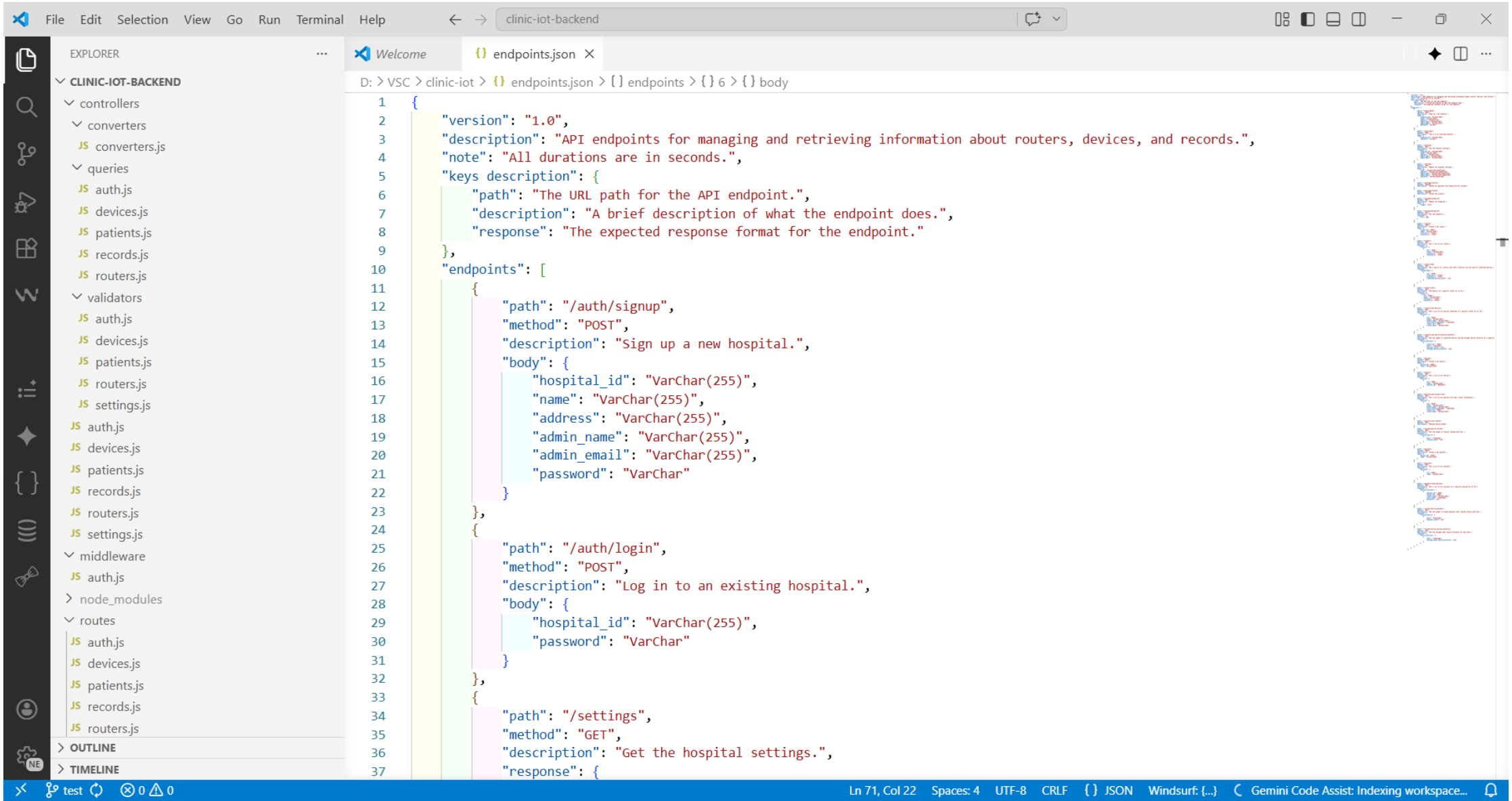
MQTT Broker

- **Eclipse Mosquitto MQTT Server:** an open source (EPL/EDL licensed) message broker that implements the MQTT protocol versions 5.0, 3.1.1 and 3.1. Mosquitto is lightweight and is suitable for use on all devices from low power single board computers to full servers.

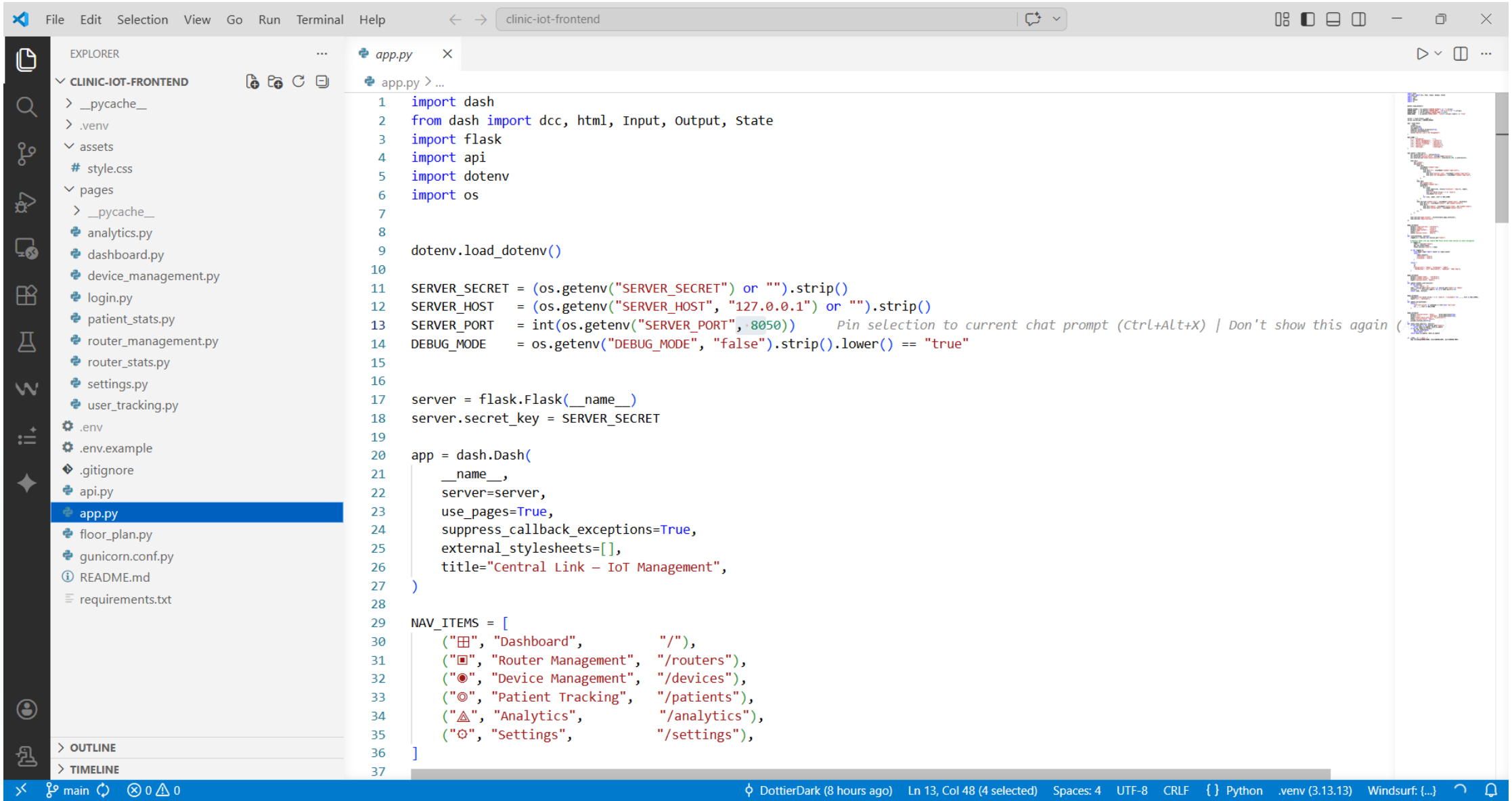
Database Setup



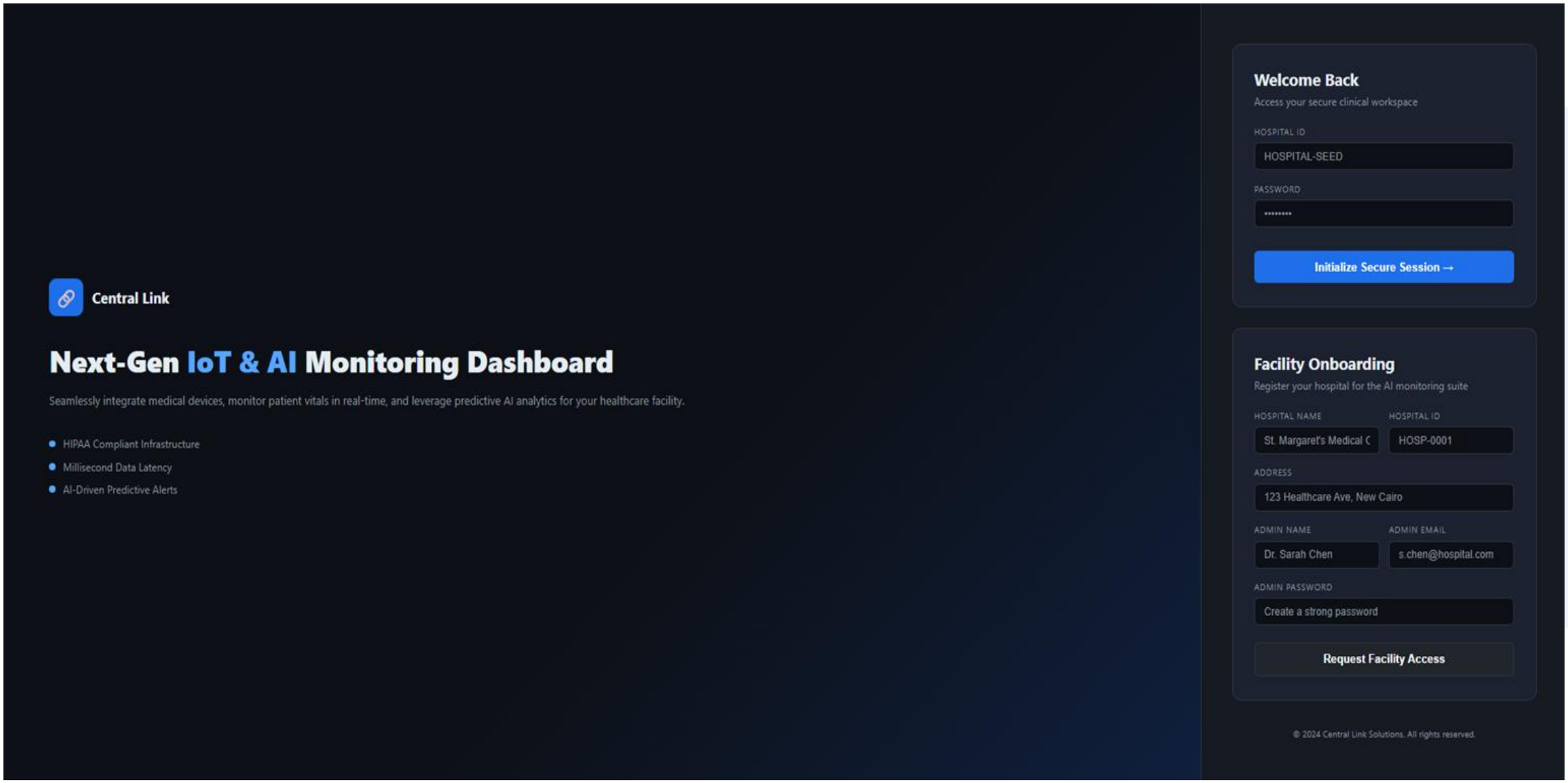
Backend


File Edit Selection View Go Run Terminal Help
clinic-iot-backend
EXPLORER
CLINIC-IOT-BACKEND
controllers
converters
converters.js
queries
auth.js
devices.js
patients.js
records.js
routers.js
validators
auth.js
devices.js
patients.js
routers.js
settings.js
auth.js
devices.js
patients.js
records.js
routers.js
settings.js
middleware
auth.js
node_modules
routes
auth.js
devices.js
patients.js
records.js
routers.js
OUTLINE
TIMELINE
Welcome
endpoints.json
D: > VSC > clinic-iot > endpoints.json > [ ] endpoints > { } 6 > { } body
{
"version": "1.0",
"description": "API endpoints for managing and retrieving information about routers, devices, and records.",
"note": "All durations are in seconds.",
"keys description": {
"path": "The URL path for the API endpoint.",
"description": "A brief description of what the endpoint does.",
"response": "The expected response format for the endpoint."
},
"endpoints": [
{
"path": "/auth/signup",
"method": "POST",
"description": "Sign up a new hospital.",
"body": {
"hospital_id": "VarChar(255)",
"name": "VarChar(255)",
"address": "VarChar(255)",
"admin_name": "VarChar(255)",
"admin_email": "VarChar(255)",
"password": "VarChar"
}
},
{
"path": "/auth/login",
"method": "POST",
"description": "Log in to an existing hospital.",
"body": {
"hospital_id": "VarChar(255)",
"password": "VarChar"
}
},
{
"path": "/settings",
"method": "GET",
"description": "Get the hospital settings.",
"response": {
test
Ln 71, Col 22
Spaces: 4
UTF-8
CRLF
{ } JSON
Windsurf: {...}
Gemini Code Assist: Indexing workspace...

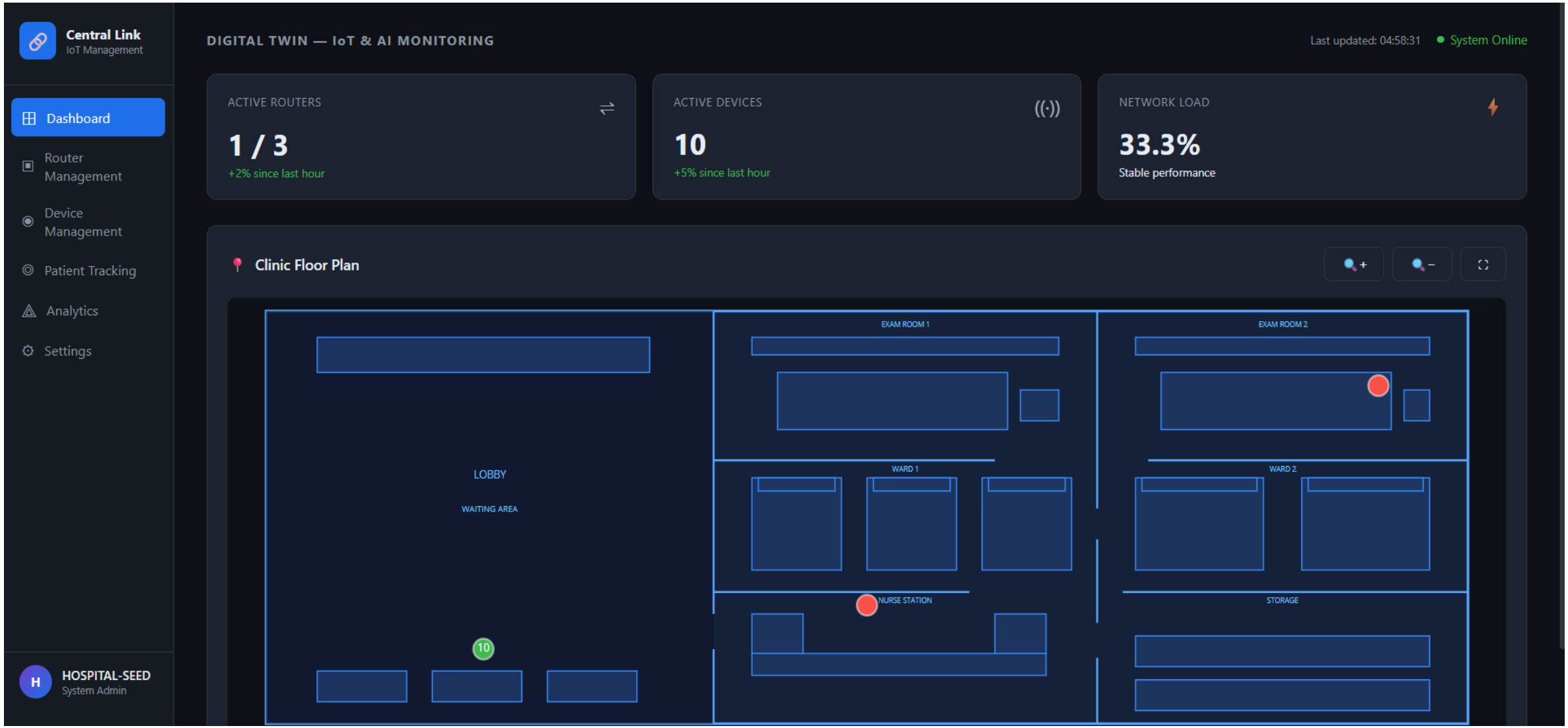
Dashboard


File Edit Selection View Go Run Terminal Help
clinic-iot-frontend
EXPLORER
CLINIC-IOT-FRONTEND
__pycache__
.venv
assets
style.css
pages
__pycache__
analytics.py
dashboard.py
device_management.py
login.py
patient_stats.py
router_management.py
router_stats.py
settings.py
user_tracking.py
.env
.env.example
.gitignore
api.py
app.py
floor_plan.py
gunicorn.conf.py
README.md
requirements.txt
OUTLINE
TIMELINE
app.py
app.py > ...
1 import dash
2 from dash import dcc, html, Input, Output, State
3 import flask
4 import api
5 import dotenv
6 import os
7
8
9 dotenv.load_dotenv()
10
11 SERVER_SECRET = (os.getenv("SERVER_SECRET") or "").strip()
12 SERVER_HOST = (os.getenv("SERVER_HOST", "127.0.0.1") or "").strip()
13 SERVER_PORT = int(os.getenv("SERVER_PORT", 8050)) Pin selection to current chat prompt (Ctrl+Alt+X) | Don't show this again (
14 DEBUG_MODE = os.getenv("DEBUG_MODE", "false").strip().lower() == "true"
15
16
17 server = flask.Flask(__name__)
18 server.secret_key = SERVER_SECRET
19
20 app = dash.Dash(
21 __name__,
22 server=server,
23 use_pages=True,
24 suppress_callback_exceptions=True,
25 external_stylesheets=[],
26 title="Central Link — IoT Management",
27)
28
29 NAV_ITEMS = [
30 ("⊞", "Dashboard", "/"),
31 ("▣", "Router Management", "/routers"),
32 ("◉", "Device Management", "/devices"),
33 ("◎", "Patient Tracking", "/patients"),
34 ("◬", "Analytics", "/analytics"),
35 ("⚙", "Settings", "/settings"),
36]
37
main 0 0
DottierDark (8 hours ago) Ln 13, Col 48 (4 selected) Spaces: 4 UTF-8 CRLF { } Python .venv (3.13.13) Windsurf: {...}

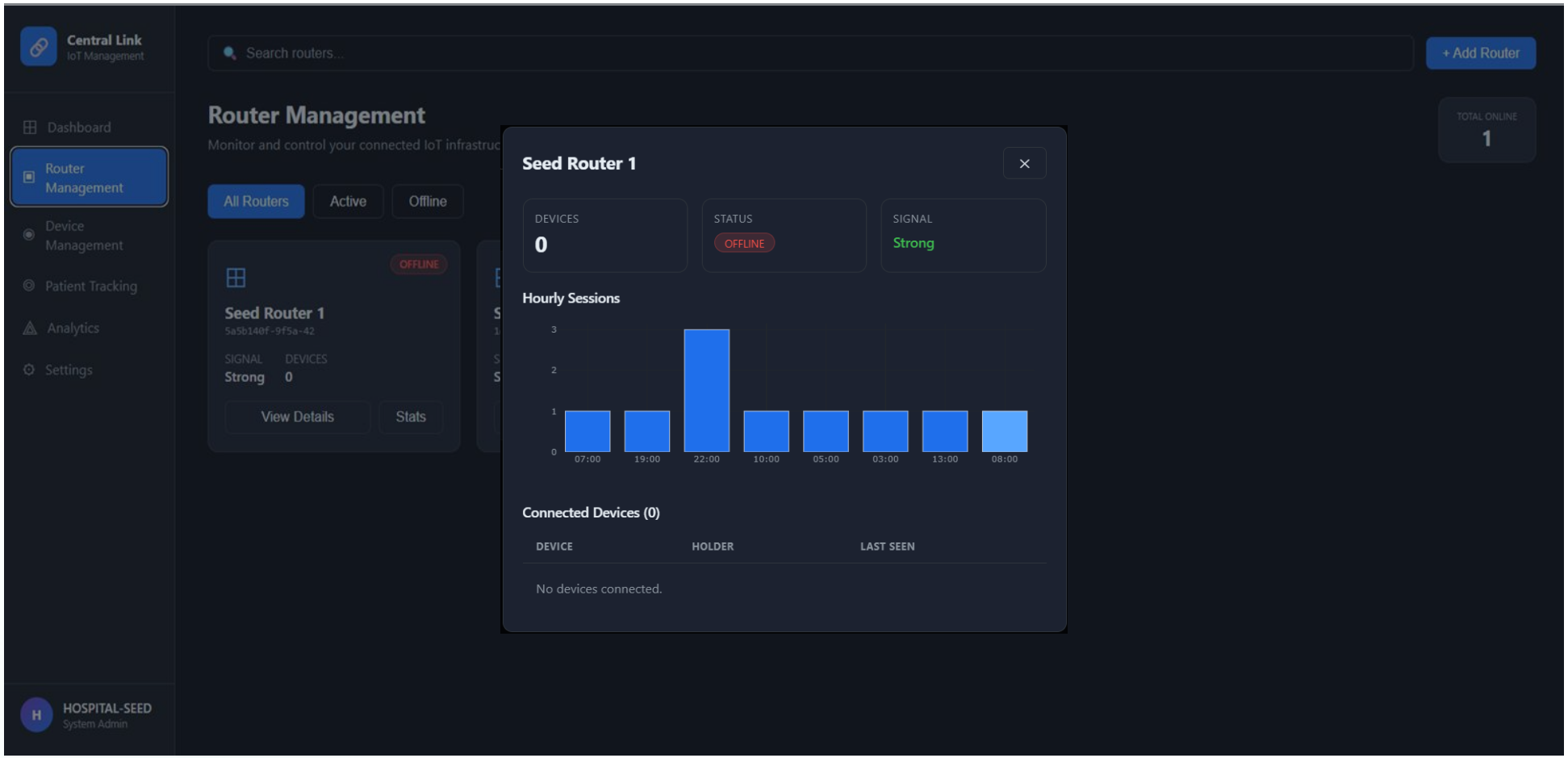
SECURE AUTHENTICATION



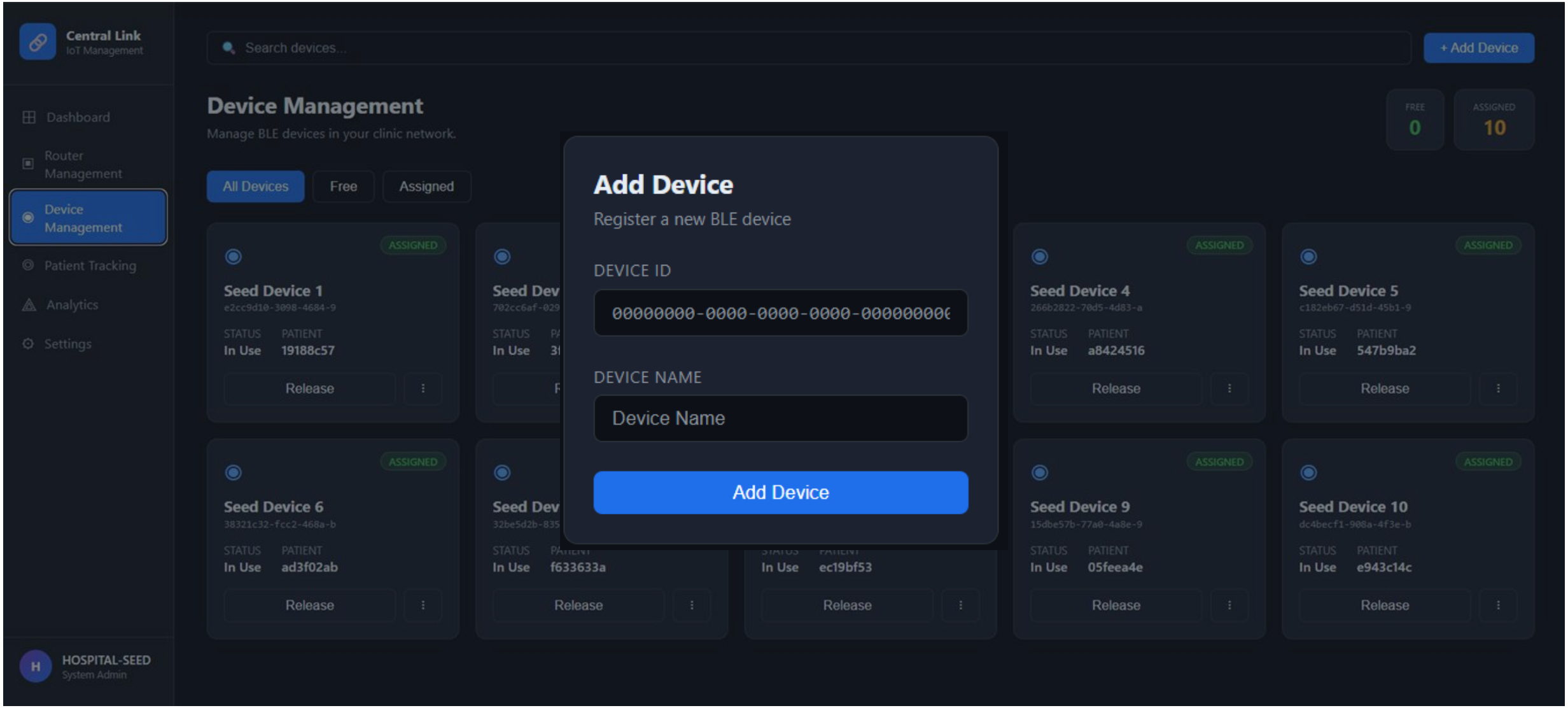
THE LIVE DASHBOARD



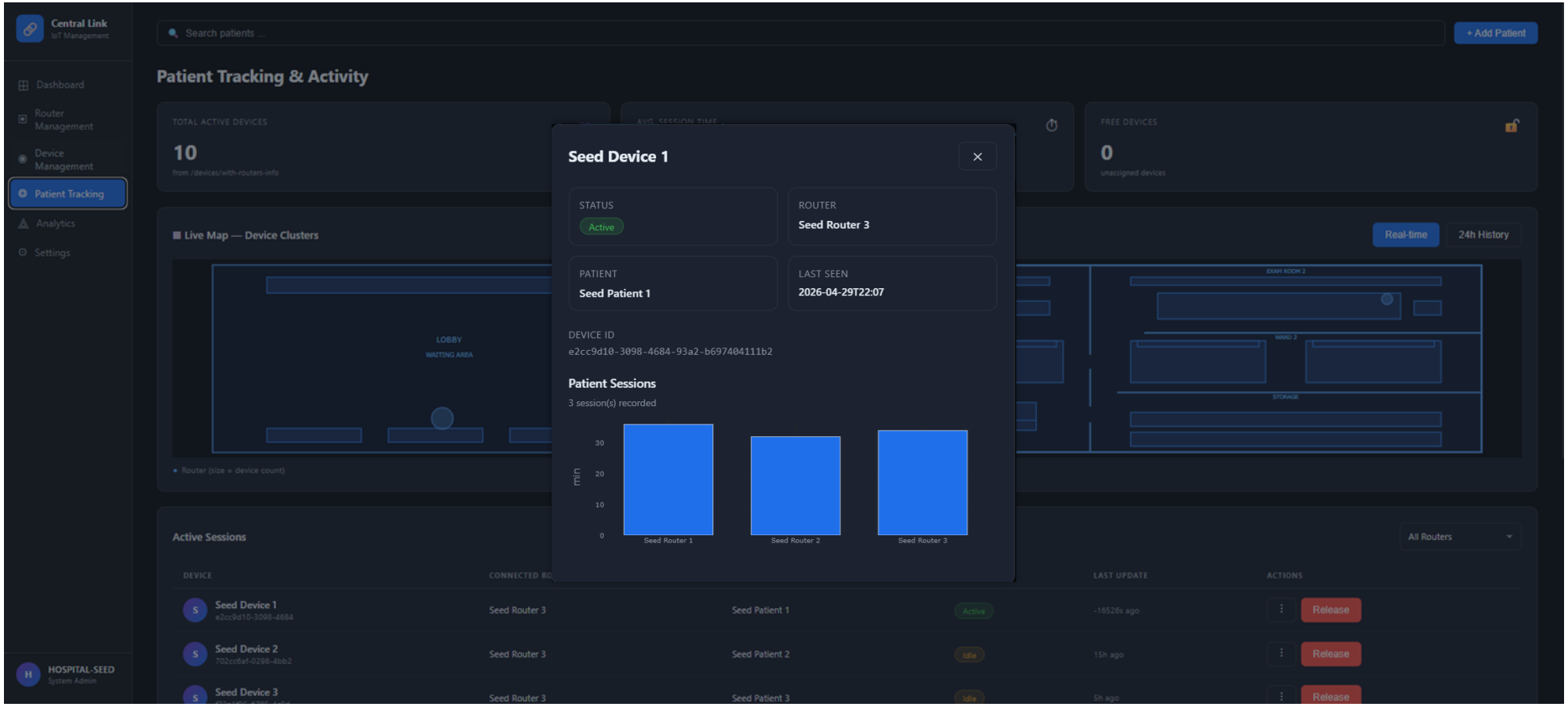
INFRASTRUCTURE (ROUTER MANAGEMENT)

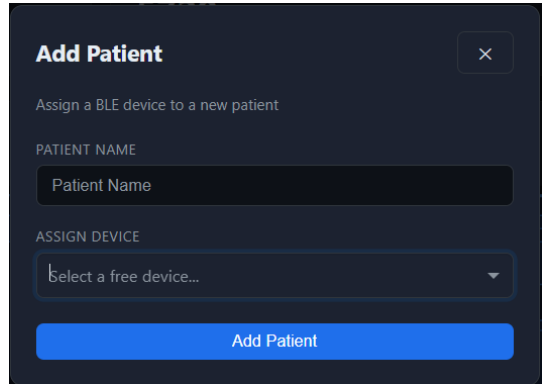


HARDWARE PROVISIONING

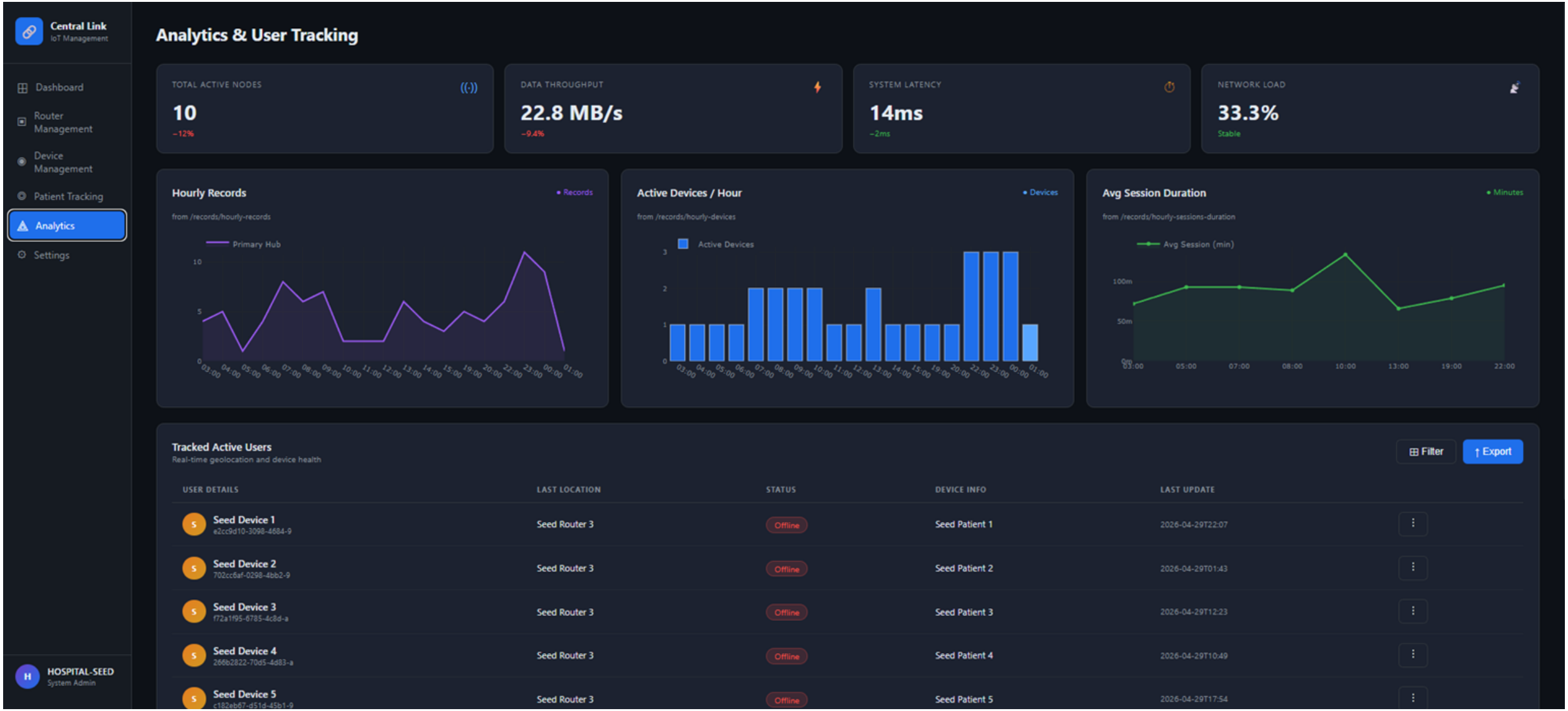


REAL-TIME PATIENT TRACKING

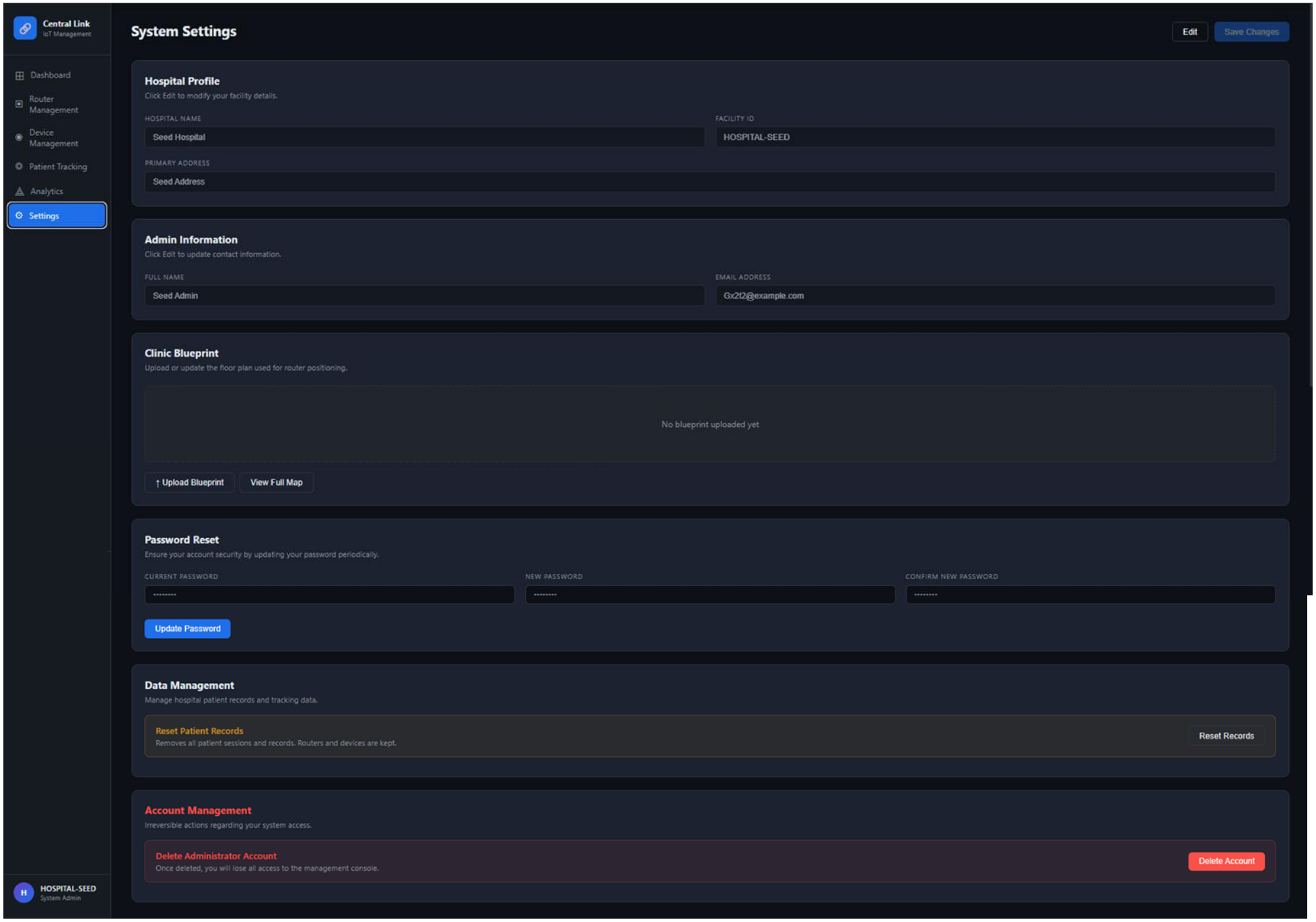




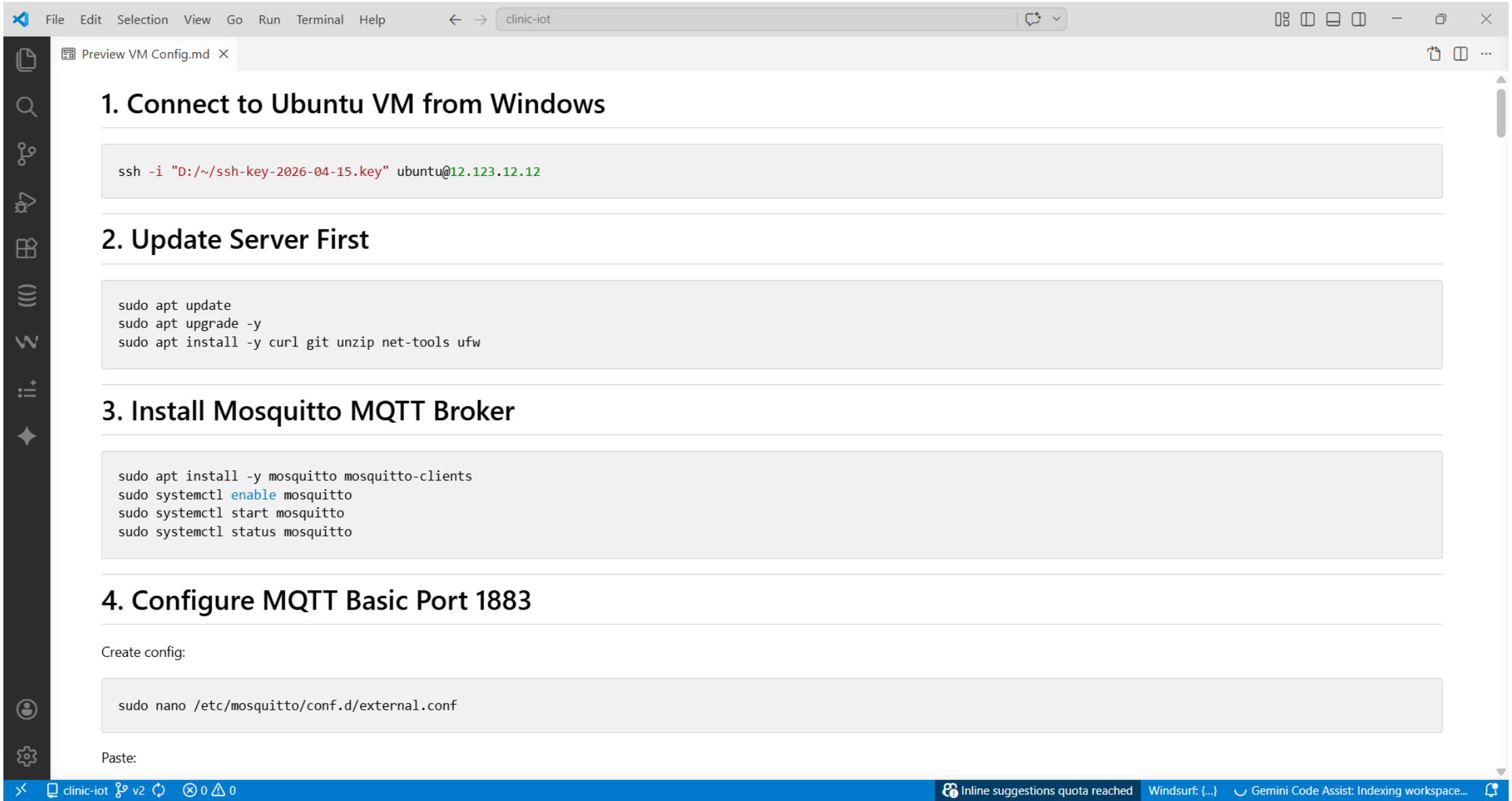
WORKFLOW ANALYTICS



SYSTEM SETTINGS



VM Setup


File Edit Selection View Go Run Terminal Help
clinic-iot
Preview VM Config.md
1. Connect to Ubuntu VM from Windows
ssh -i "D:/~/ssh-key-2026-04-15.key" ubuntu@12.123.12.12
2. Update Server First
sudo apt update
sudo apt upgrade -y
sudo apt install -y curl git unzip net-tools ufw
3. Install Mosquitto MQTT Broker
sudo apt install -y mosquitto mosquitto-clients
sudo systemctl enable mosquitto
sudo systemctl start mosquitto
sudo systemctl status mosquitto
4. Configure MQTT Basic Port 1883
Create config:
sudo nano /etc/mosquitto/conf.d/external.conf
Paste:
clinic-iot v2
0 0
Inline suggestions quota reached
Windsurf: {...}
Gemini Code Assist: Indexing workspace...

Domain

https://central-link-iot.duckdns.org

CHALLENGES & FUTURE WORK

Technical Hurdles

Signal Multipathing: Indoor noise creates RSSI fluctuations.

Latency: Maintaining sub-100ms updates.

Final Deliverables

ML Anomaly Module: Training anomaly detection models to flag unexpected patient loops.

Scale Testing: Validating system performance with 20+ active tags.

ACADEMIC REFERENCES

- **Arumugam, G. R., & Muthaiyah, S. (2024).** A Proof of Concept Using BLE to Optimize Patients' Turnaround Time... *IJISRT.*]
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Thank You!

Questions & Discussion

Central Link Team | Medical Clinic Digital Twin

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